

Stage 1 Offline Phase: Cost-Benefit Modeling

Historical Logs

Data preparation process



Record original historical data

id	Pipeline	Execution e	Cost	Perf.
1	$OP_1 \rightarrow OP_2 \rightarrow \dots \rightarrow OP_K$	[0.3, 0.6, 1.0, ...]	320	0.74
2	$OP_1 \rightarrow OP_3 \rightarrow \dots \rightarrow OP_K$	[0.5, 1.0, 0.4, ...]	195	0.69
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
N	$OP_2 \rightarrow OP_1 \rightarrow \dots \rightarrow OP_K$	[1.0, 0.7, 1.0, ...]	450	0.81

Operator selection & Pipeline logic

execution degree of operator OP_i

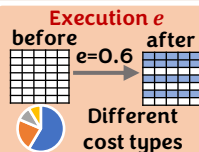
actual cost under execution e

downstream model performance

Convert to Training Data

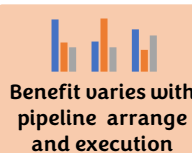
a. Cost Model Training Data

Operator i	Execution e	Cost c
OP_1	[0.1, 0.2, ..., 1.0]	[21, 45, ..., 230]
OP_2	[0.1, 0.2, ..., 1.0]	[33, 39, ..., 64]
$\vdots$	$\vdots$	$\vdots$
OP_K	[0.1, 0.2, ..., 1.0]	[84, 98, ..., 179]



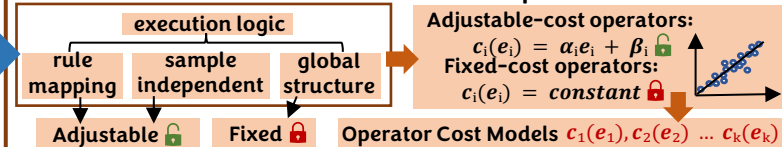
b. Benefit Model Training Data

Encoded Pipe	Execution e	Dataset feat. d	Model feat. m	Perf.
$\varphi(P^1)$	[0.3, 0.6, 1.0, ...]	d^1	m^1	0.74
$\varphi(P^2)$	[0.5, 1.0, 0.4, ...]	d^1	m^1	0.69
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$\varphi(P^N)$	[1.0, 0.7, 1.0, ...]	d^r	m^s	0.81

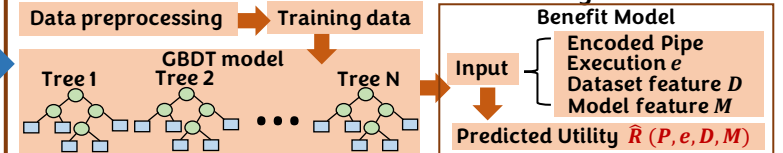


Learn Models

a. Cost Model for Each Operator



b. Benefit Model: GBDT Surrogate

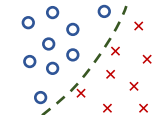


Stage 2 Online Phase: Pipeline Optimization

INPUTS

1					
2					
3					
4					
5					
$\vdots$					
N					

Original data



Downstream model

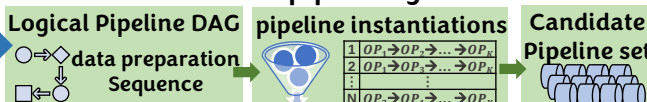


User demand

① User Demand Synthesis

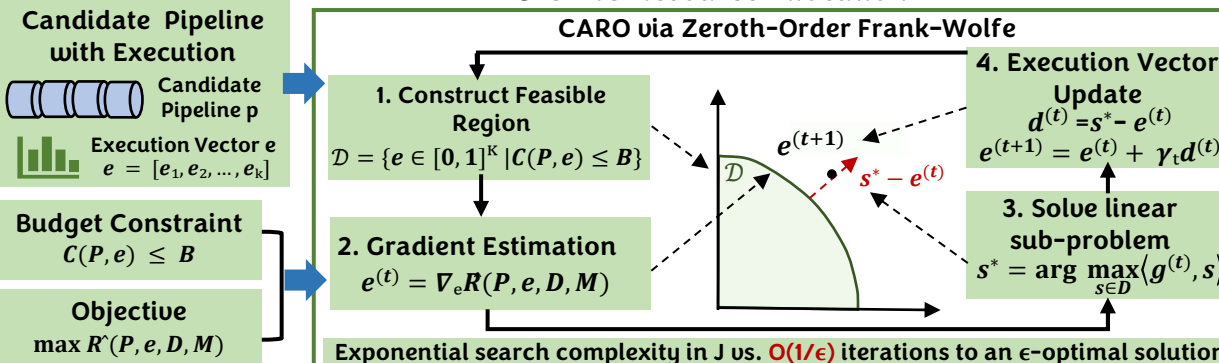


② Candidate pipeline generation

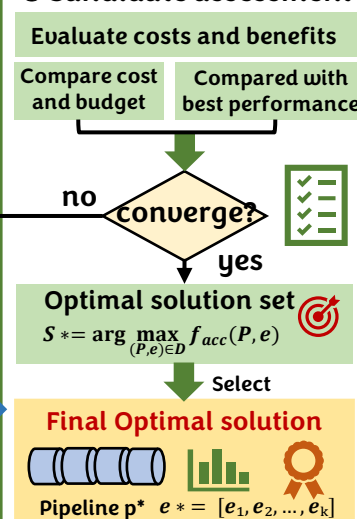


③ CARO Resource Allocation

CARO via Zeroth-Order Frank-Wolfe



④ Candidate assessment



OUTPUTS

1					
2					
3					
4					
5					
$\vdots$					
N					

Clean data



Low cost



User satisfaction